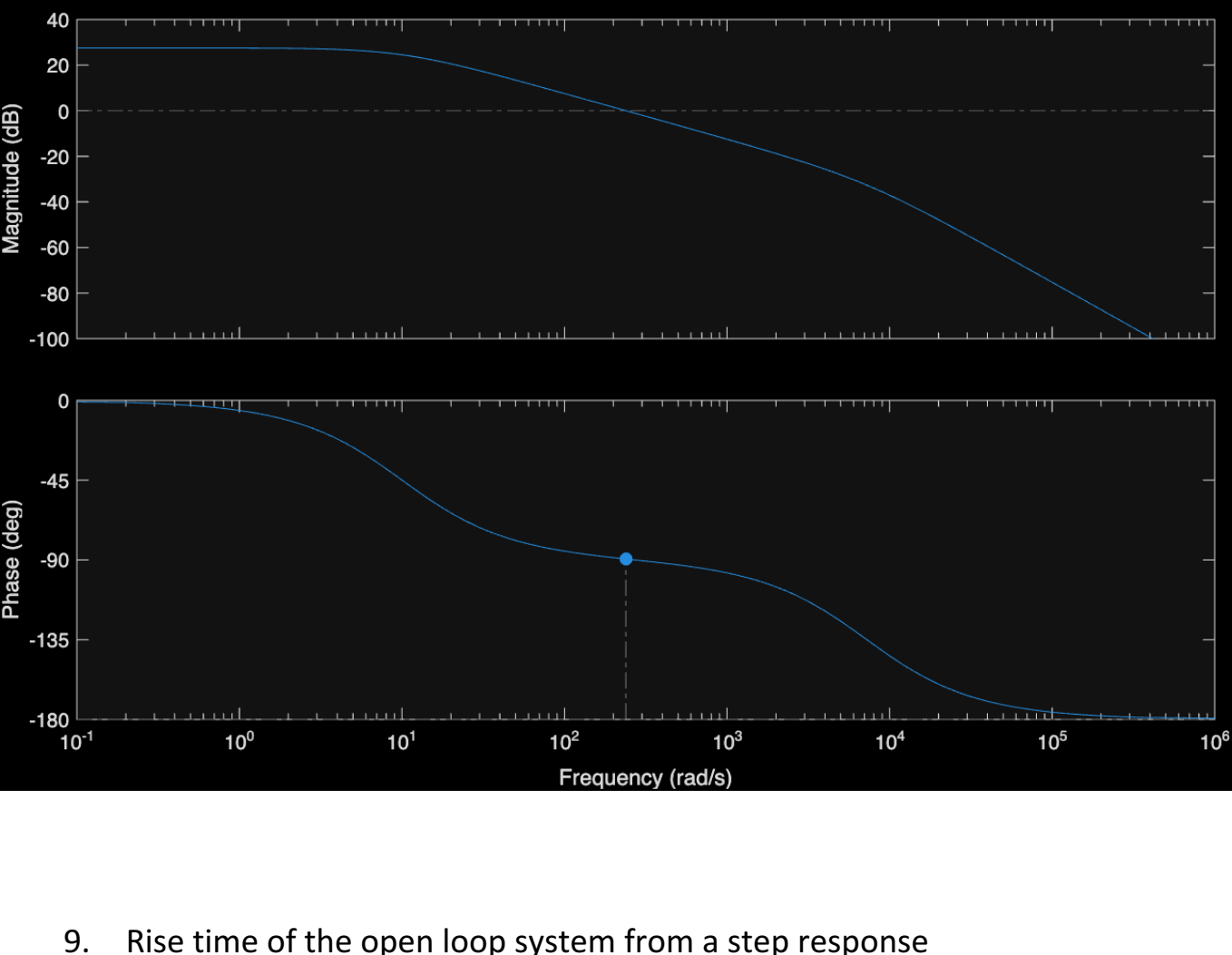


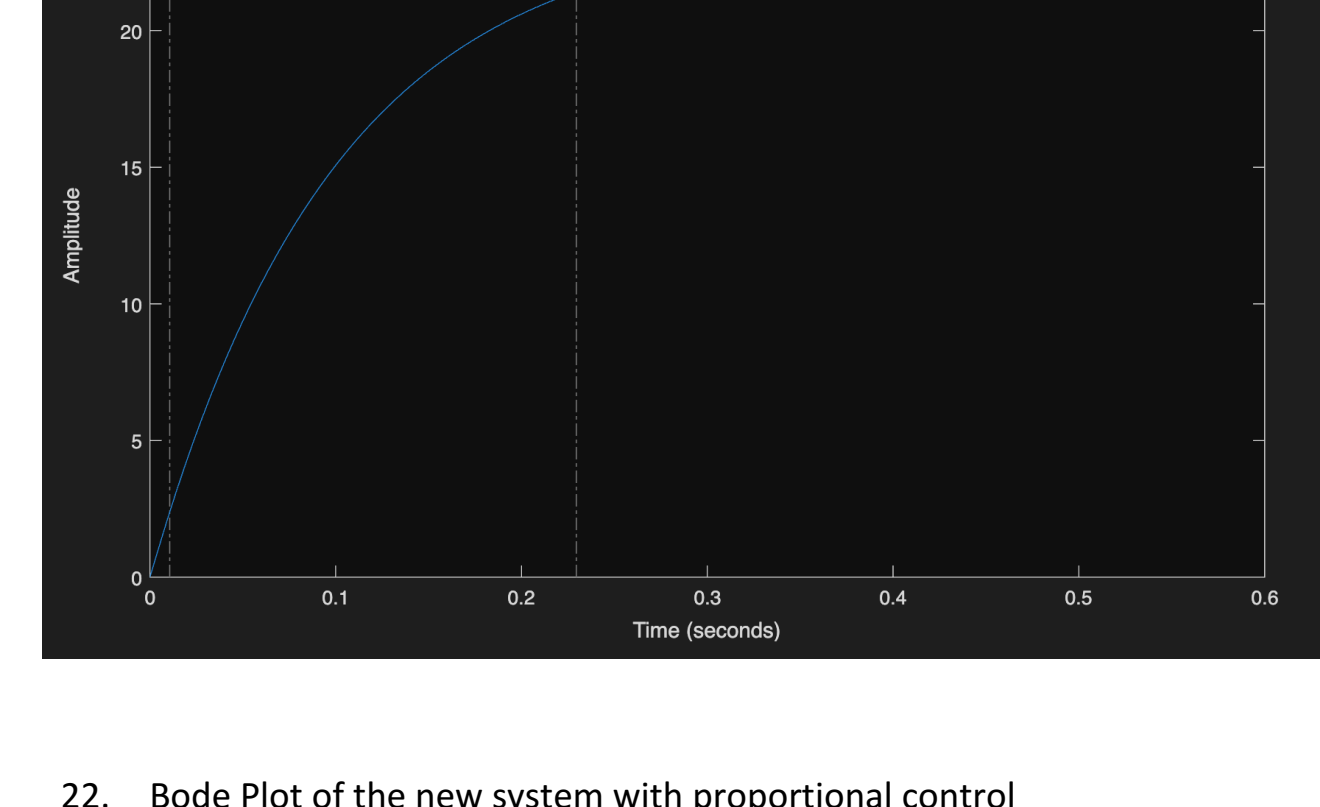
Final Qube Design

11/21/25 10:22 AM

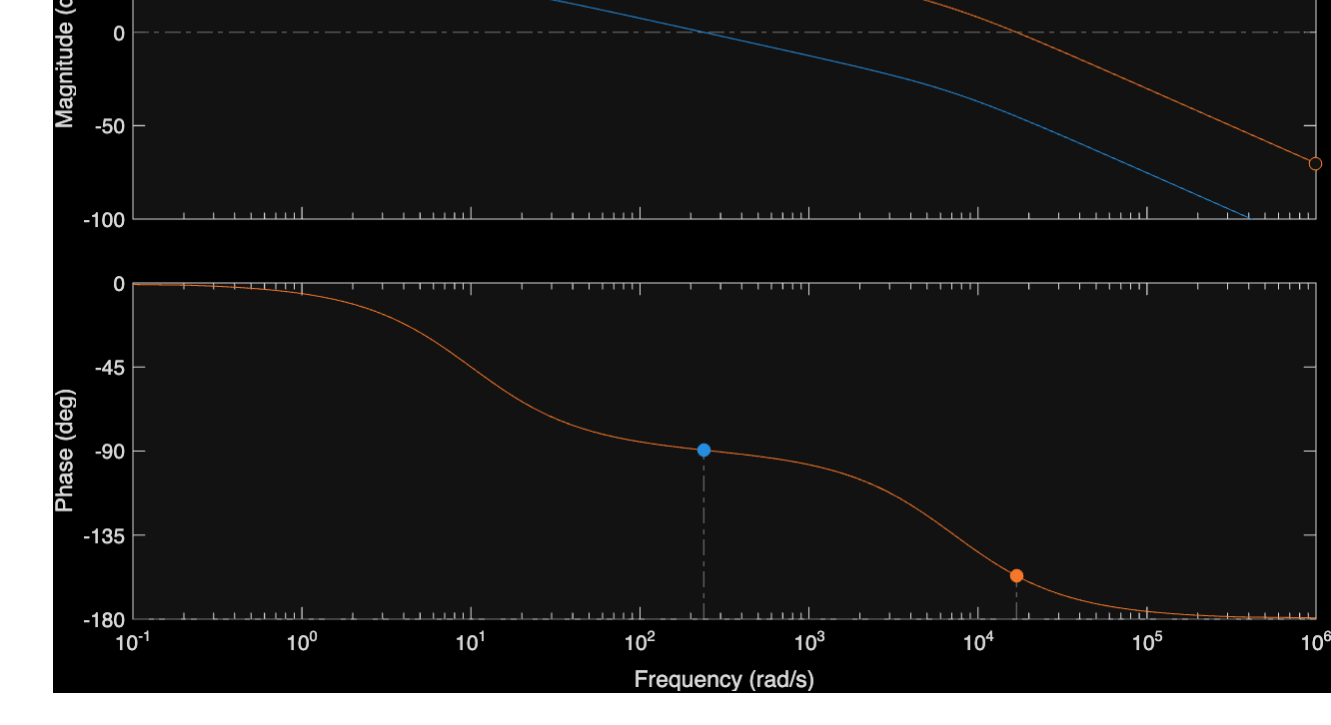
1. Bode Diagram of uncompensated system



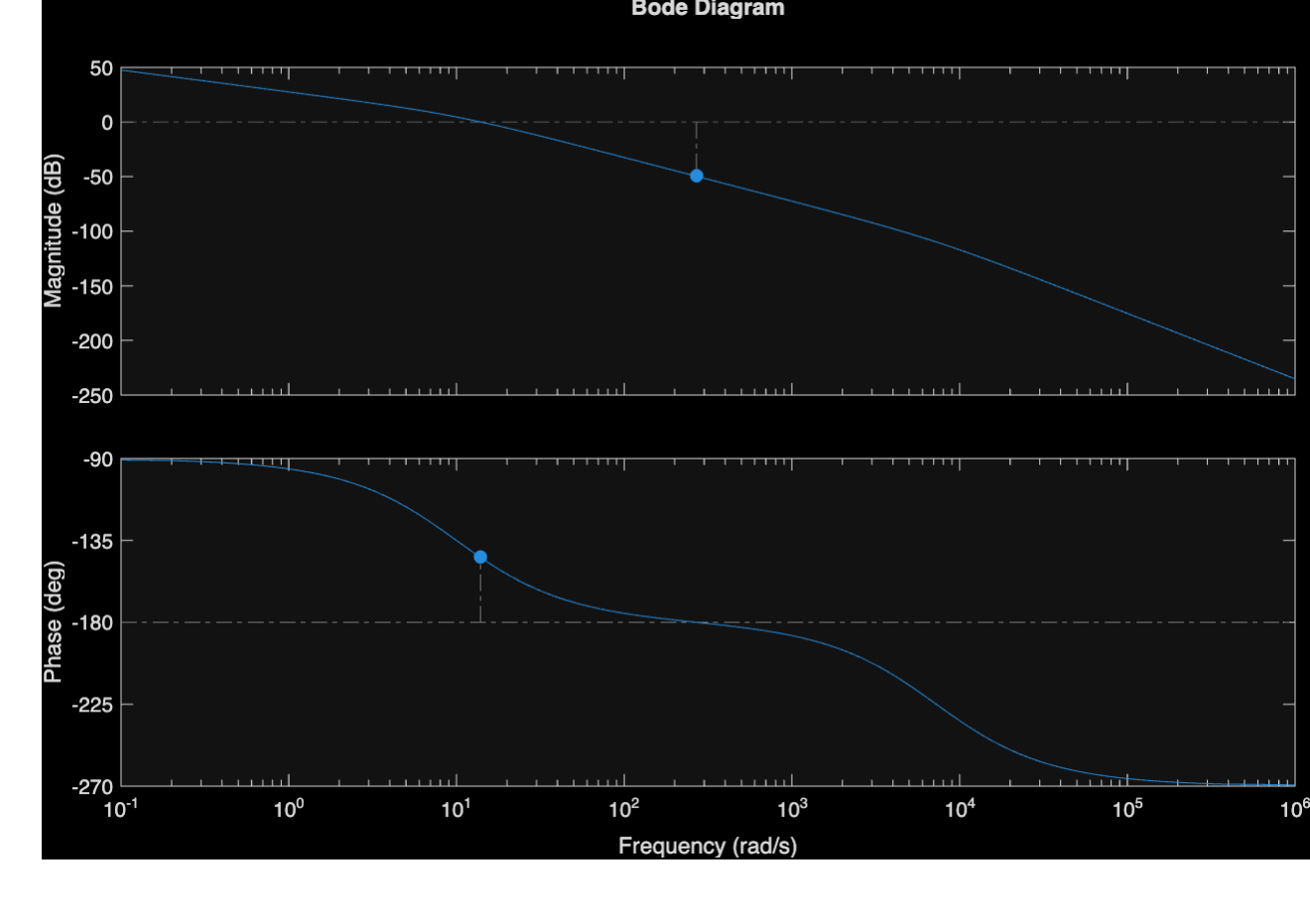
9. Rise time of the open loop system from a step response



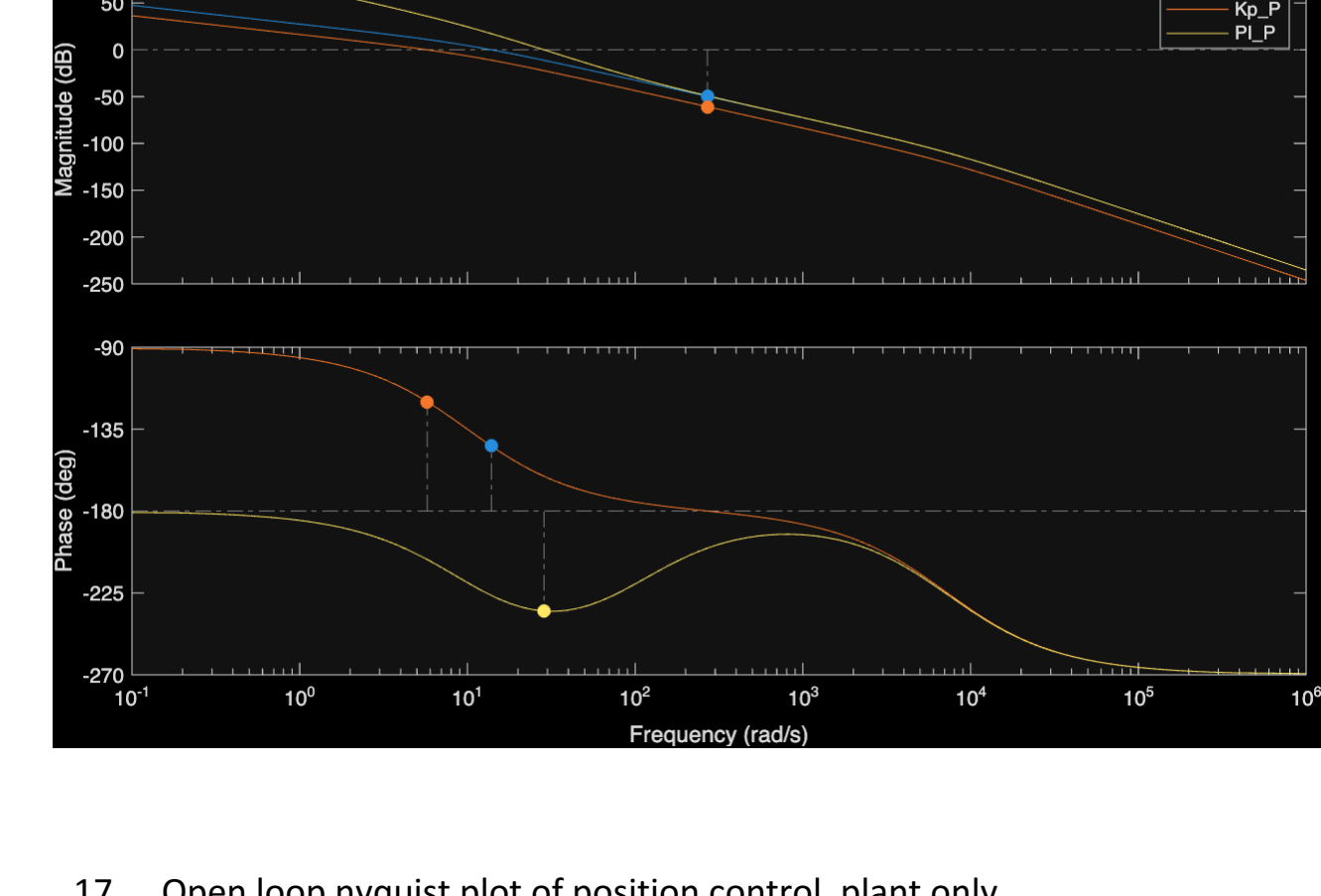
22. Bode Plot of the new system with proportional control



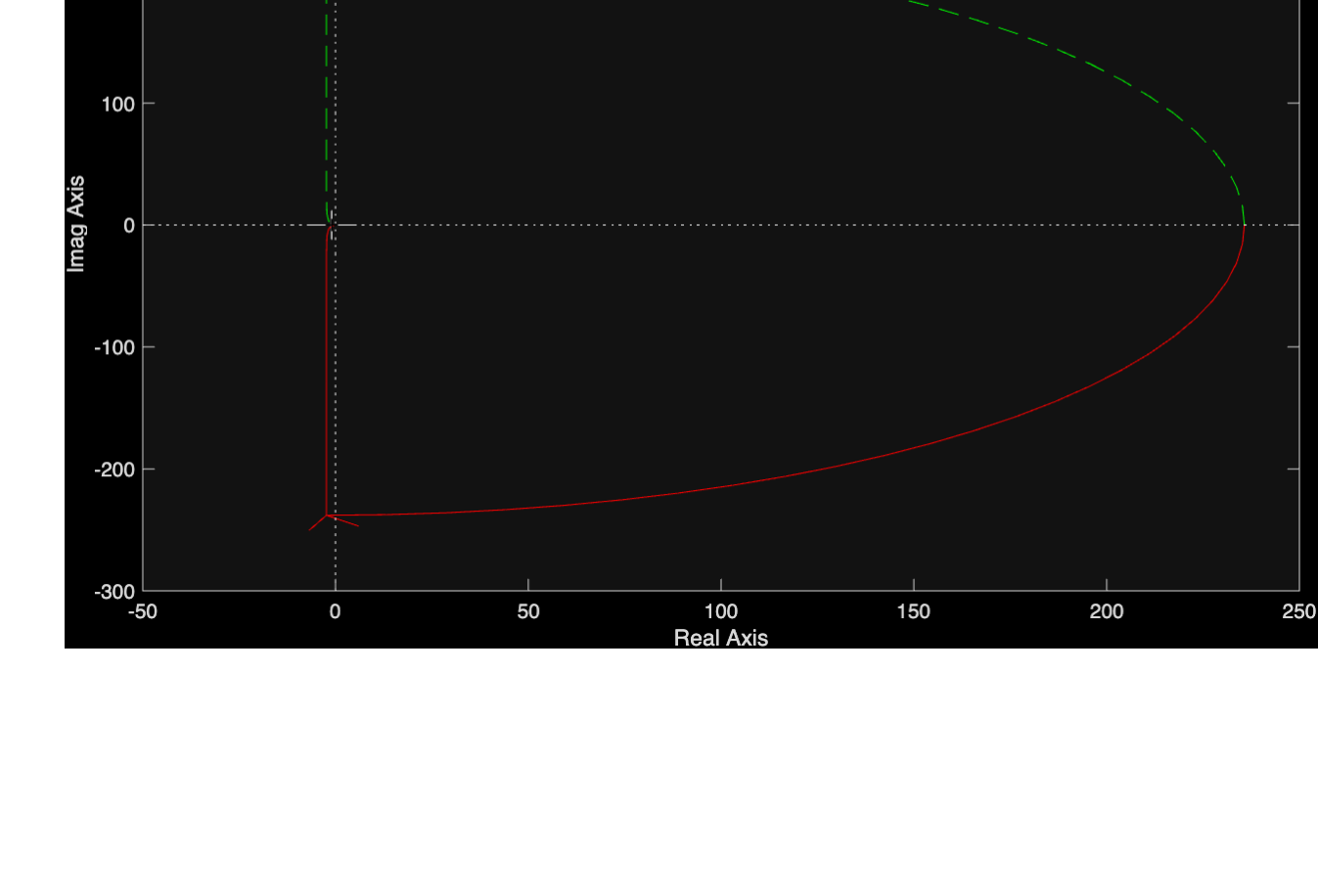
16. The open loop position control bode plot



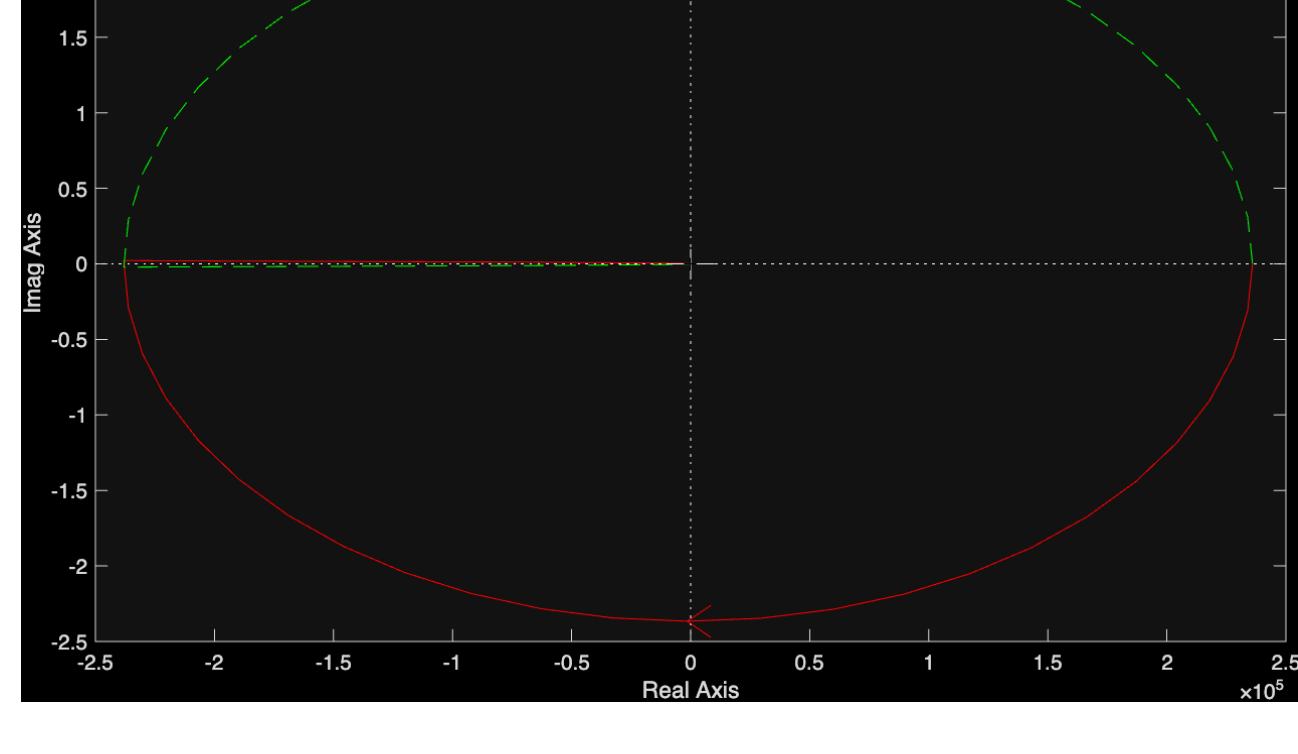
22. P and PI control on the bode plot. Blue = uncomp. Red = P control Yellow = PI Control



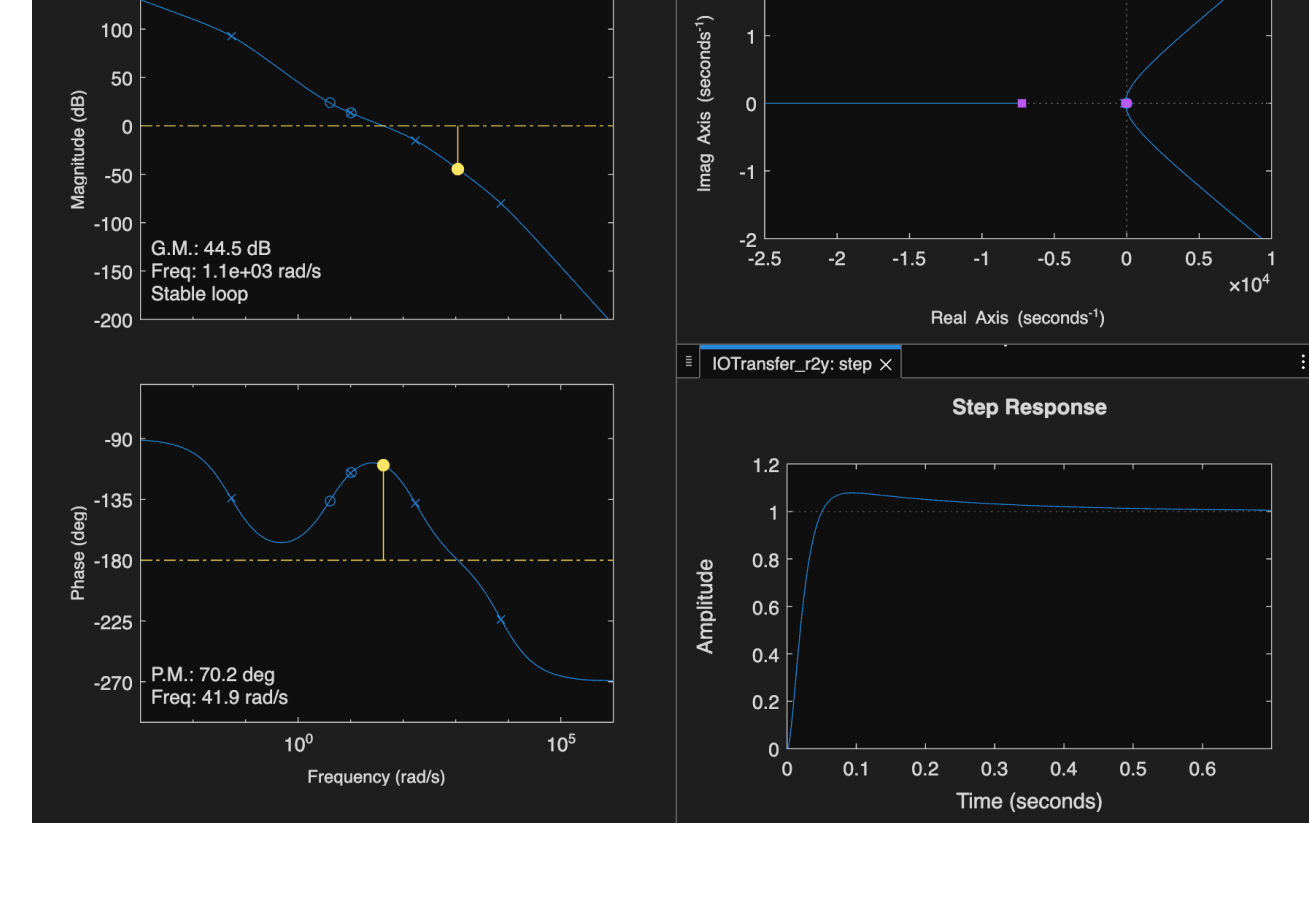
17. Open loop nyquist plot of position control, plant only



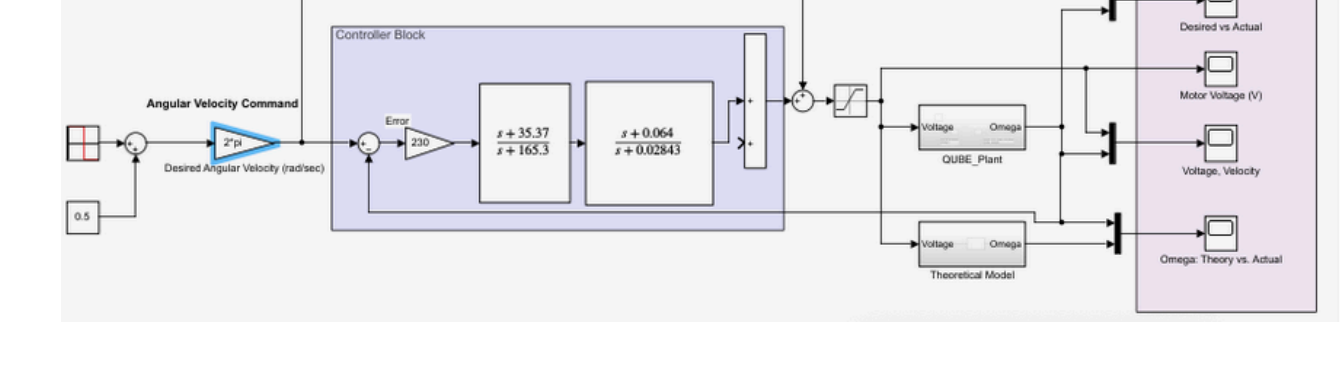
24. Open loop nyquist plot of PI controller motor system servo position control



31. Control system view of my full lead AND lag compensator design



33. Simulink block of final compensator design (I forgot to take a screenshot with my original design)



34. Final Compensator Response to unit step (first showing rise time/overshoot and second showing steady-state behavior, third is response to a unit disturbance input

